

Batch-Binding Time from 5 to 60 Minutes Does Not Limit Capture of His-CasP from Clarified Lysate

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ABSTRACT

Capture of His-CasP on nickel resin was compared after 5, 20, and 60 min of batch mixing. Bound target recovery was 73%, 75%, and 74%, respectively, and flow-through target remained 9-11% of the load. Extending contact time did not improve capture. In a separate control, reducing resin volume from 1.0 to 0.25 mL decreased recovery to 38%, showing that resin capacity rather than binding time was limiting under the low-resin condition. The study supports a 5-min batch step at the tested load.

Keywords: batch binding, Ni-NTA, residence time, His-CasP, negative result

INTRODUCTION

Long batch-binding times are often adopted without evidence that equilibrium is slow. When resin capacity is sufficient, most capture may occur within minutes.

This study separated contact time from resin amount. The primary experiment used the same resin volume at three mixing times. A low-resin control was included to verify that the assay could detect capacity limitation.

MATERIALS AND METHODS

Lysate and resin

Clarified lysate containing 4.0 mg of His-CasP was mixed with 1.0 mL of 50% nickel-resin slurry in 20 mM phosphate, pH 7.8, 300 mM NaCl, and 10 mM imidazole. Mixing was performed at 8 °C.

Time comparison

Independent samples were mixed for 5, 20, or 60 min. Resin was collected by low-speed centrifugation, washed, and eluted with 300 mM imidazole.

Capacity control

A separate sample used 0.25 mL slurry for 20 min. Bound, flow-through, and wash fractions were quantified by densitometry against a purified standard.

Group	Slurry volume	Binding time
T5	1.0 mL	5 min
T20	1.0 mL	20 min
T60	1.0 mL	60 min
Low resin	0.25 mL	20 min

Table 1. Contact-time and capacity conditions.

RESULTS

Contact time did not affect recovery

Bound His-CasP recovery was 73 +/- 4% at 5 min, 75 +/- 3% at 20 min, and 74 +/- 5% at 60 min. Flow-through target was 11 +/- 2%, 9 +/- 2%, and 10 +/- 3%, respectively. The small differences lacked a monotonic trend.

Low resin revealed capacity limitation

At 0.25 mL slurry, bound recovery fell to 38 +/- 6% and flow-through target increased to 45 +/- 5%. This result confirmed that the assay was sensitive to insufficient resin capacity.

Process implication

Because 5 min matched 60 min at the standard resin amount, the longer batch step added time without improving capture.

Condition	Bound recovery	Flow-through
5 min, 1.0 mL	73 +/- 4%	11 +/- 2%
20 min, 1.0 mL	75 +/- 3%	9 +/- 2%
60 min, 1.0 mL	74 +/- 5%	10 +/- 3%
20 min, 0.25 mL	38 +/- 6%	45 +/- 5%

Table 2. Binding time had little effect; low resin capacity reduced capture.

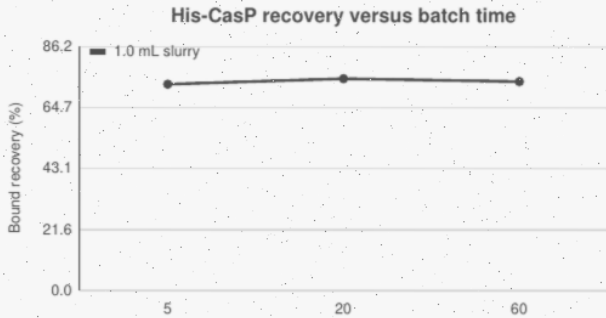


Figure 1. Recovery plateaued by the first measured time point.

DISCUSSION

The standard resin amount captured His-CasP within 5 min. Extending mixing to 20 or 60 min did not increase recovery.

The low-resin control demonstrates that capture can fail when capacity is insufficient. Therefore, the negative time result should not be generalized to arbitrary protein loads or resin amounts.

LIMITATIONS

Times shorter than 5 min were not tested. The result applies to the stated load, buffer, temperature, and slurry geometry.

CONCLUSION

A 5-min batch-binding step was sufficient for His-CasP at 1.0 mL slurry. Longer contact did not improve recovery, whereas reducing resin volume exposed a clear capacity limitation.

REFERENCES

The following references are fictional and are included only to test reference-section handling.

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